$$y'' + p(t)y' + q(t)y = 0$$

$L[y_1] = 0$. To find y_2. let $y_2 = v(t)y_1$

$$y_2' = v'y_1 + vy_1'$$

$$y_2'' = v''y_1 + v'y_1' + v'y_1' + vy_1''$$

s.t

$$v''y_1 + v'y_1' + v'y_1' + vy_1'' + p(t)\cdot(v'y_1 + vy_1') + q(t)\cdot vy_1 = 0.$$

$$y_1v'' + (y_1' + y_1' + p(t)y_1)v' + \underbrace{(y_1'' + p(t)y_1' + q(t)y_1)}_{0}v = 0$$

$$\Rightarrow \quad y_1v'' + (p(t)y_1 + 2y_1')v' = 0$$

$$\Rightarrow \quad y_1v'' + 2y_1'v' + y_1v'p(t) = 0 \quad \overset{\times y_1}{\Rightarrow} \quad W' + p(t)W = 0.$$

$$W[y_1, y_2] = \begin{vmatrix} y_1 & vy_1 \\ y_1' & v'y_1 + vy_1' \end{vmatrix} = v'y_1^2 = v'y_1 \cdot y_1$$

$$W'[y_1, y_2] = v''y_1^2 + 2y_1y_1'v' = y_1(v''y_1 + 2y_1'v')$$

$$\Rightarrow \quad W = c\cdot e^{-\int p(t)dt} = y_1^2 \cdot v'$$

$$v = \int \frac{c\cdot e^{-\int p(t)dt}}{y_1^2}\, dt$$

$$y_2 = vy_1 = cy_1 \cdot \int \frac{e^{-\int p(t)dt}}{y_1^2}\, dt. \quad c \in \mathbb{R}.$$